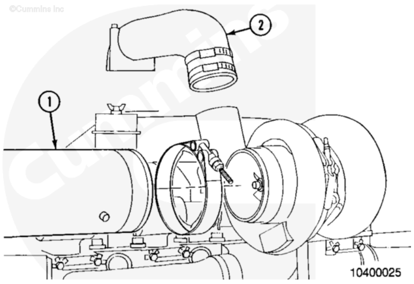
Trainings ▾

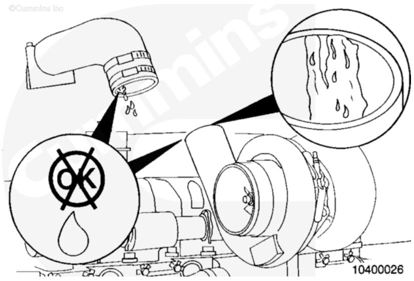
Initial Check

Remove the air intake pipe (1) and air crossover (2) from the turbocharger.

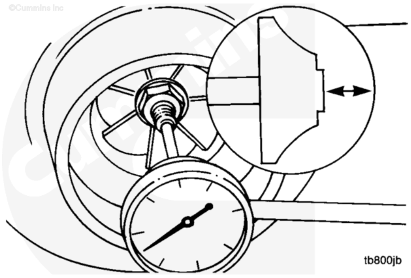
Remove and discard the turbocharger V-band clamps.



Examine the compressor discharge and air crossover piping for oil.

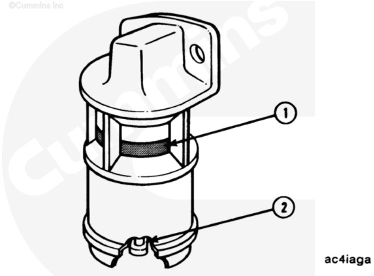


If oil is visible, check the turbocharger axial and radial clearance contained in this procedure.

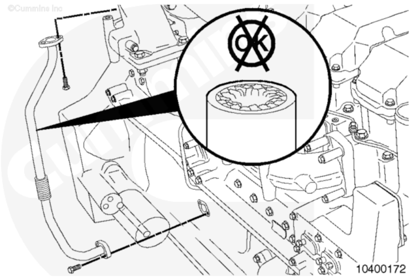


Check for intake restriction. Refer to Procedure 010-031 in Section 10 (/qs3/pubsys2/xml/en/procedures/89/89-010-031.html).

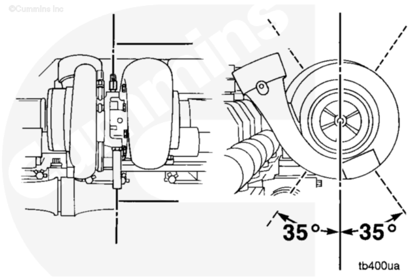
If the restriction exceeds specifications, replace or clean the air filter element. Refer to equipment manufacturer instructions.



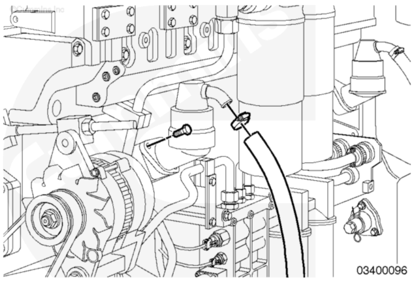
Remove the oil drain tube and check for restrictions. Clear any restrictions that are found.



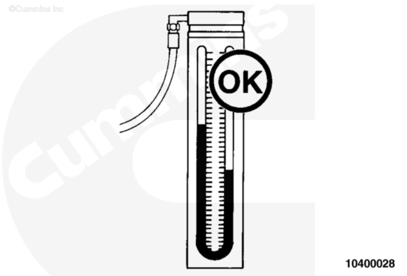
Check the angle of the drain tube. The angle of the tube **must** be within 35 degrees of vertical. Adjust the turbocharger, if necessary.



If the drain tube is free of restrictions and at the correct angle, check the crankcase breathers and tubes to be sure they are **not** plugged.



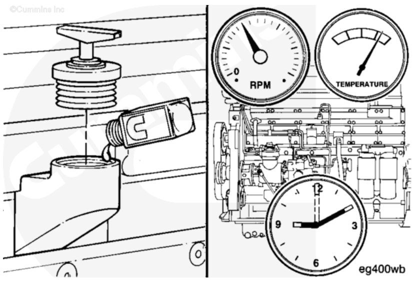
If these checks do **not** reveal the problem, measure the crankcase pressure (blowby). Refer to Procedure 014-005 in Section 14 (/qs3/pubsys2/xml/en/procedures/89/89-014-005.html).



The following is the procedure for the fluorescent dye tracer test:

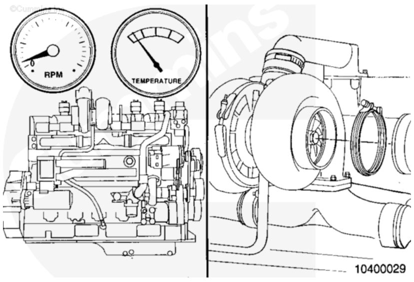
Add one unit of fluorescent tracer to each 38 liters [10 gal] of engine lubricating oil.

Operate the engine at low idle for 10 minutes.



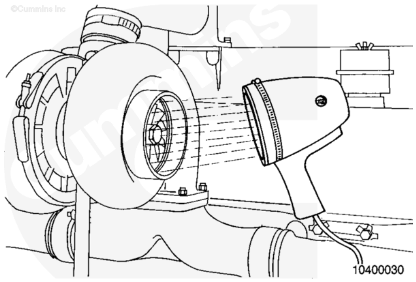
Shut the engine OFF.

Allow the turbocharger to cool and remove the exhaust pipe from the turbine housing.

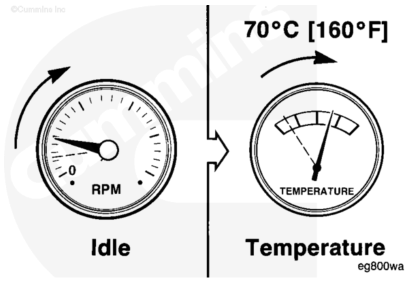


Use a high intensity black light to inspect the turbine housing outlet for oil.

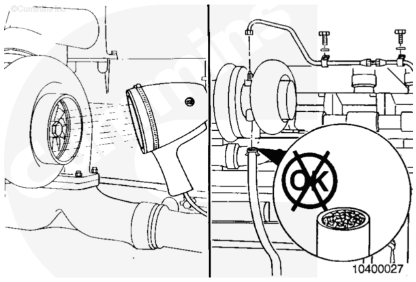
A dark blue glow indicates oil and a yellow glow indicates fuel.



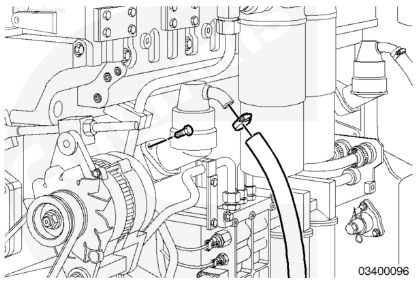
If fuel is found in the turbine housing, the problem is excess, unburned fuel. To correct the problem, decrease the idle period or increase the idle rpm. The coolant temperature needs to be maintained at a minimum of 71°C [160°F].



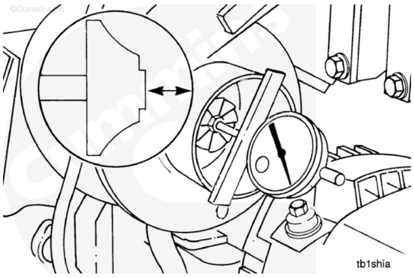
If oil is found in the turbine housing, remove the oil drain tube and check for restriction. Clear any restrictions found.



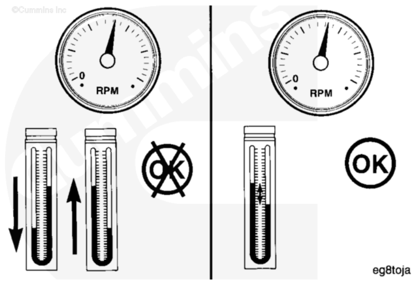
If the drain tube is free of restrictions and at the correct angle, check the crankcase breather(s) and tube(s) to be sure they are **not** plugged.



Check the turbocharger axial clearance and radial clearance contained in this procedure.



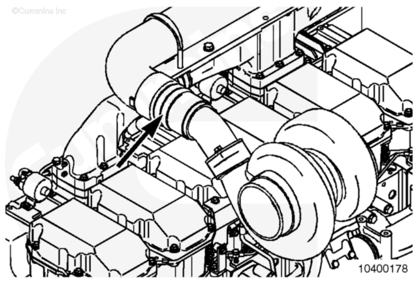
If these checks do **not** reveal the problem, measure the crankcase pressure (blowby). Refer to Procedure 014-005 in Section 14 (/qs3/pubsys2/xml/en/procedures/89/89-014-005.html).



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove and discard the turbocharger V-band clamps.

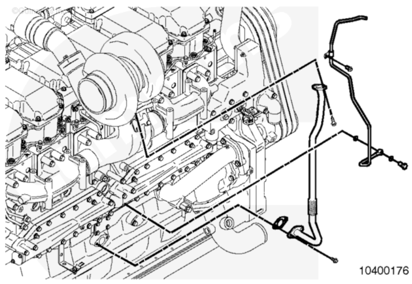
Remove the coupling from the turbocharger compressor outlet.

Remove any original equipment manufacturer (OEM) supplied exhaust piping.

Remove the turbocharger oil drain line.

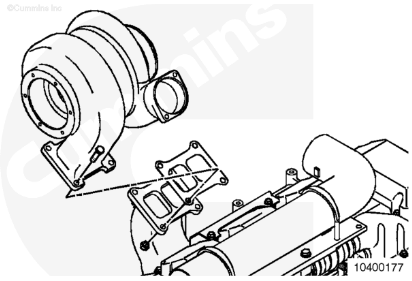
Remove the turbocharger oil supply line and line clamps.

Note : The turbocharger oil supply line can not be totally removed from the engine without removing the exhaust manifold. It is acceptable to remove all mounting hardware and allow the oil supply line to hang while removing and installing the turbocharger.



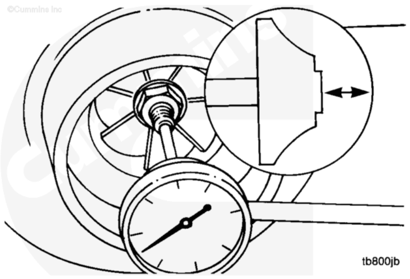
Remove the turbocharger mounting nuts and bolts.

Remove the turbocharger and gasket.



Measure

Use dial depth gauge, Part Number ST-537, or a dial indicator to measure the axial motion (end-to-end).

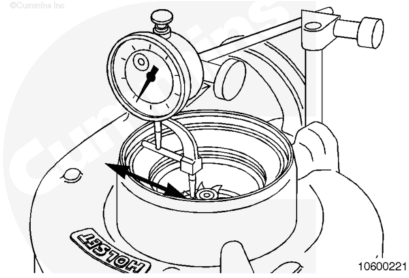


Compressor Impeller Axial End Clearance			
	mm		in
Holset® HX82 Turbocharger	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HX60 Turbocharger	0.051	MIN	0.002
	0.152	MAX	0.006
Holset® HX83 Turbocharger	0.025	MIN	0.001
	0.152	MAX	0.006
Schwitzer S500	0.07	MIN	0.003
	0.18	MAX	0.007

Make sure the movement is within MIN/MAX total indicator reading values shown above.

Note : If the end clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Measure the radial clearance (side-to-side) at the compressor impeller nose using a dial gauge.

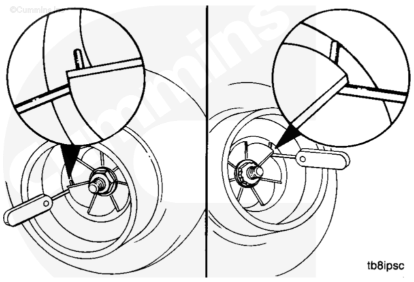


Compressor Impeller Radial End Clearance			
	mm		in
Holset® HX82 Turbocharger	0.254	MIN	0.01
	0.787	MAX	0.031
Holset® HX83 Turbocharger	0.254	MIN	0.01
	0.787	MAX	0.031
Holset® HX60 Turbocharger	0.356	MIN	0.014
	0.635	MAX	0.025

Make sure the movement is within MIN/MAX total indicator reading values shown above.

Note : If the radial clearance does not meet the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Use a wire-type feeler gauge to measure the radial clearance (side-to-side).



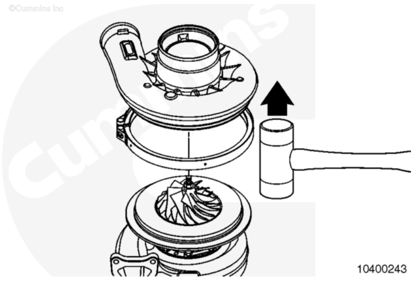
Schwitzer S500 Radial Clearance		
mm		in
0.5	MIN	0.02
0.92	MAX	0.36

Note : If the radial clearance does not meet the specifications, the turbocharger must be replaced or rebuilt.

Disassemble

⚠ **CAUTION** ⚠

The compressor blades can be easily damaged with the compressor cover is removed.

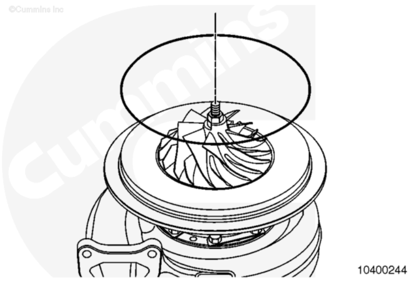


Loosen and remove the V-band clamp securing the compressor housing.

Note : Some ratings use turbochargers with a titanium compressor wheel. The edge of the wheel is very sharp. Handle the compressor wheel with care.

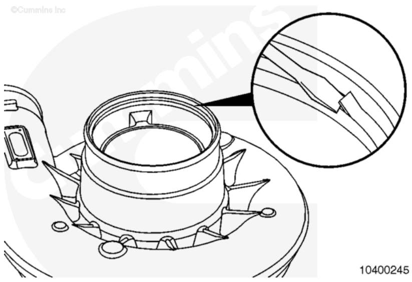
Use a mallet to gently tap the compressor cover to remove the compressor cover from the turbocharger.

Remove and discard the o-ring.



⚠ **CAUTION** ⚠

The compressor blades can be easily damaged with the compressor cover is removed.

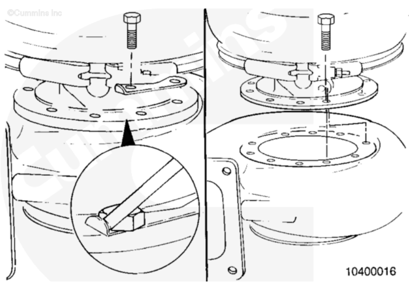


If the compressor cover has an inlet baffle, it can be removed with a flat tip screwdriver.

Pry the retaining ring up and remove the retaining ring.

⚠ **CAUTION** ⚠

The compressor blades can be easily damaged with the compressor cover is removed.

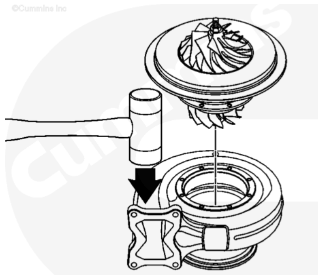


Remove the turbine housing by bending the lock plate tabs away from the capscrew heads.

Remove the capscrews.

Make sure the turbine wheel rotates freely inside the turbine housing.

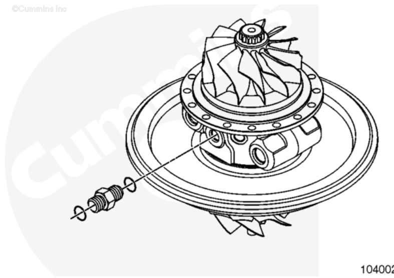
Tap on the turbine housing with a mallet to separate the turbine housing from the bearing housing.



10400246

Remove the oil supply fitting from the center housing assembly.

Discard the o-rings



10400247

Clean and Inspect for Reuse

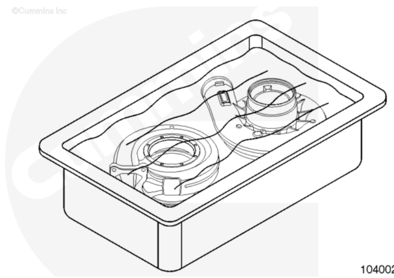
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the compressor cover and turbine housing in parts cleaner to loosen deposits.

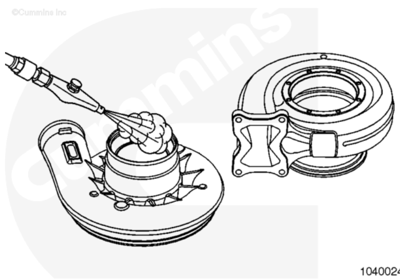


10400248

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

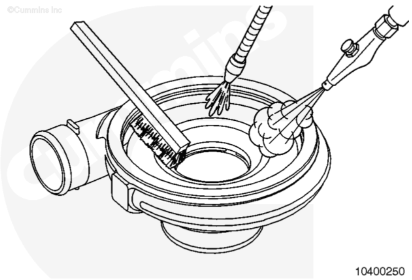
Dry the compressor housing with compressed air.



10400249

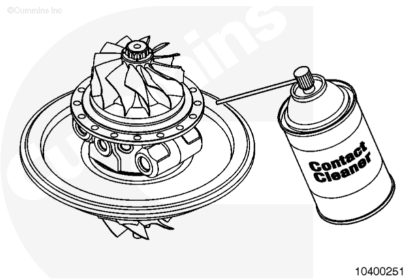
Heavy scale deposits can be removed using a soft bristle brush.

Wash and dry the parts after brushing.



⚠ CAUTION ⚠

Do not use compressed air to dry or spin the turbine wheel as this can cause damage to the turbocharger bearing system.



If chemical and brush cleaning is **not** effective, bead blast with plastic abrasive media, Part Number 3822735, or equivalent.

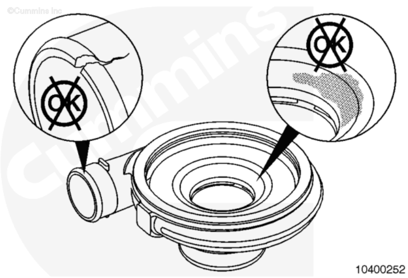
Wash and dry the parts after bead blasting.

The center housing rotating assembly can be cleaned with electrical contact cleaner, Part Number 3824510, or equivalent.

Do **not** spray contact cleaner into the oil and coolant ports.

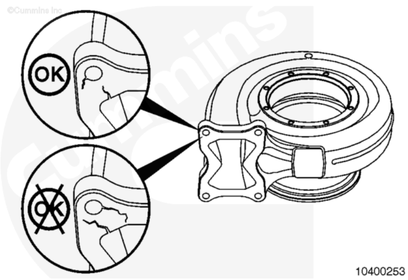
Inspect the compressor cover for cracks, dents, and other damage.

Inspect inside of the compressor cover for signs of scoring or damage from compressor wheel. Replace the compressor cover, if damaged.



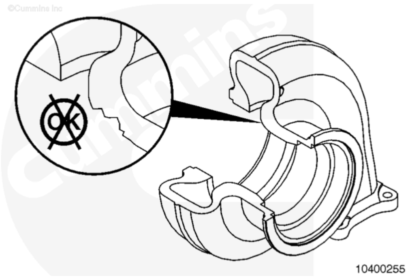
Inspect the turbine housing for cracks and other damage. Cracks larger than 10 mm [0.39 in] can cause an exhaust leak. Replace the turbine housing, if damage is found.

Follow the criteria shown in the illustration.



Inspect the turbine housing for signs of contact with the turbine wheel.

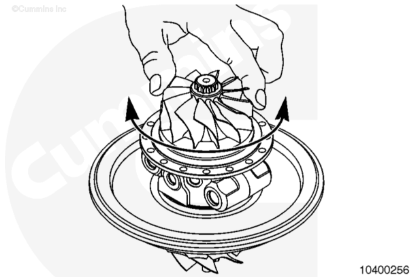
Damaged turbine housings **must** be replaced.



Inspect the center housing rotating assembly for any signs of cracks, damaged threads, or other damage.

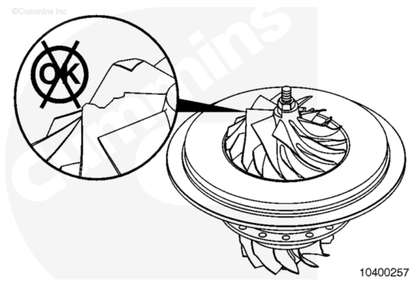
Rotate the turbine wheel by hand to make sure it spins freely.

Replace the center housing rotating assembly if it is damaged or does **not** spin freely.



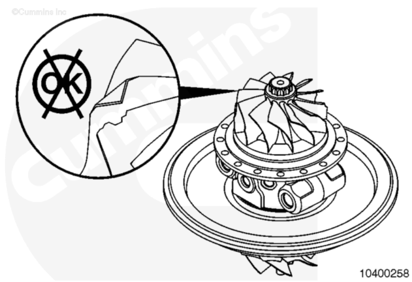
Inspect the compressor wheel for damaged, broken, or bent blades.

Do **not** attempt to repair or straighten the blades; instead replace the center housing rotating assembly.



Inspect the turbine wheel for damaged, broken, or bent blades.

Do **not** attempt to repair or straighten the blades; instead replace the center housing rotating assembly.

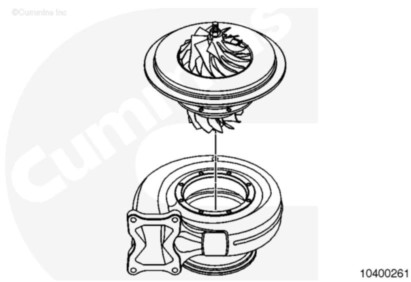


Assemble

⚠ CAUTION ⚠

The turbine blades can be easily damaged when installing the assembly in the turbine housing.

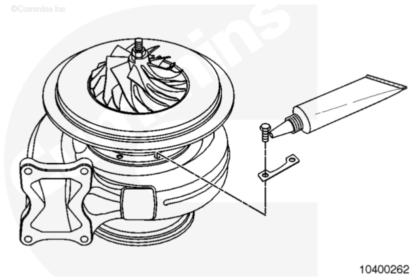
Assemble the center housing rotating assembly onto the turbine housing.



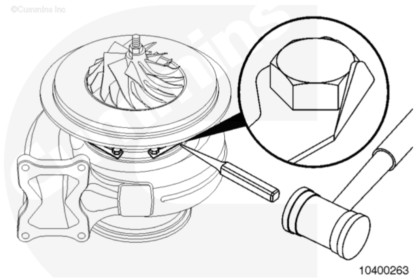
Lubricate the threads of the new capscrews with anti-seize compound, Part Number 3824732, or equivalent.

Install the capscrews, with new lock plates, to the turbocharger turbine housing.

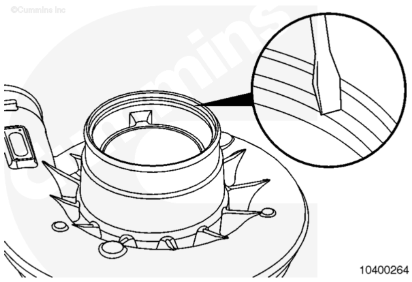
Torque Value: 20.3 n•m [177 in-lb]



Use a hammer and punch to fold the locking plate onto the flat of the capscrew, to prevent loosening.

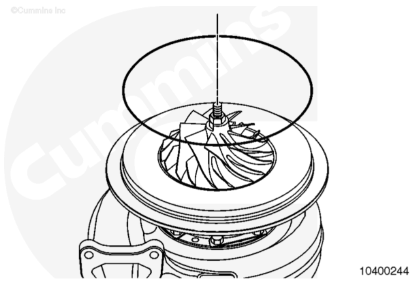


Assemble the inlet baffle into the compressor cover and secure with a retaining ring. Hold one end of the retaining ring in the groove. Press the remainder of the retaining ring into position. Use a flat tip screwdriver to make sure the retaining ring is correctly seated in the compressor cover groove.



Note : Some ratings use turbochargers with a titanium compressor wheel. The edge of the wheel is very sharp. Handle the compressor wheel with care.

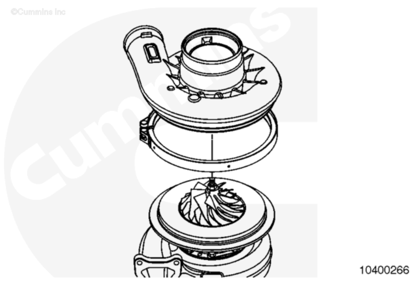
Assemble a new o-ring onto compressor housing on the turbocharger.



⚠ CAUTION ⚠

Do not damage the compressor wheel blades during assembly.

Assemble the V-band clamp and compressor cover onto the turbocharger.



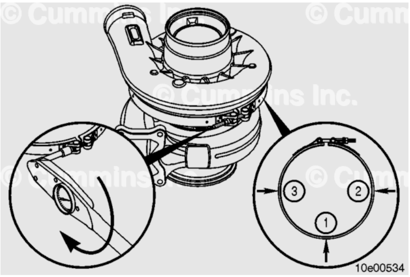
Install V-band nut and make sure everything is aligned correctly.

Do **not** use any lubrication on the V-band nut or thread.

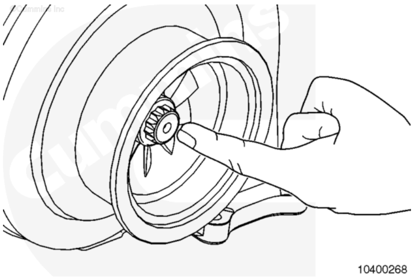
Position the compressor housing in the desired orientation.

Tighten V-band clamp.

Torque Value: 12 n•m [106 in-lb]



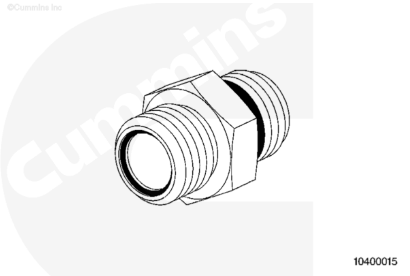
Make sure the compressor wheel and turbine wheel rotate freely.



Note : All turbocharger fittings are the flat face o-ring type fittings. Make sure the o-ring is in place before attaching the hose(s) to the fittings.

The Cummins Turbo Technologies™ turbocharger uses a 9/16-18 UNF straight thread, o-ring type oil supply fitting.

Torque Value: 45 n•m [33 ft-lb]



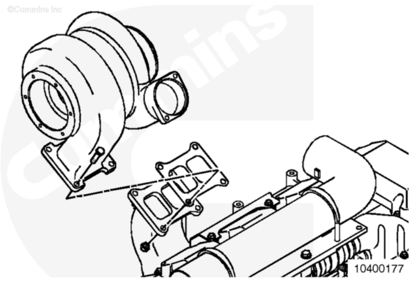
Install

Apply a film of high-temperature anti-seize compound to the turbocharger capscrew threads.

Install the turbocharger, gasket, capscrews, and nuts onto the exhaust manifold.

Tighten the nuts and capscrews.

Torque Value: 50 n•m [37 ft-lb]

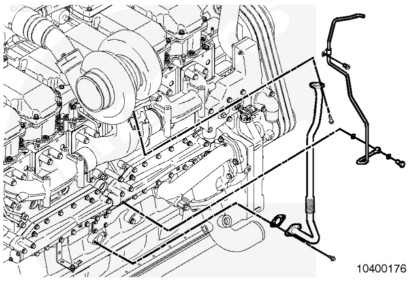


Install the turbocharger oil drain tube, gaskets, and capscrews. The angle of the tube **must** be within 35 degrees of vertical.

Tighten the drain tube capscrews.

Torque Value: 66 n•m [49 ft-lb]

Install the turbocharger oil supply line, gaskets, and line clamps.



Tighten the oil supply fitting capscrews to the turbocharger.

Torque Value: 66 n•m [49 ft-lb]

Tighten the banjo fitting fastener to the cylinder block.

Torque Value: 30 n•m [22 ft-lb]

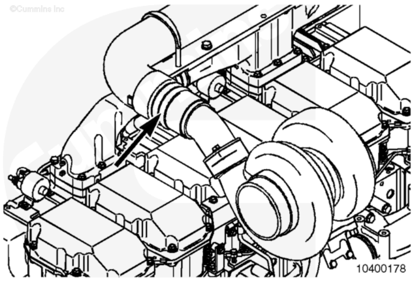
Apply a thin film of high temperature anti-seize compound to the exhaust elbow capscrew threads.

Install any OEM-supplied exhaust piping.

Install the coupling onto the turbocharger compressor outlet.

Install and tighten the new turbocharger V-band clamps.

Torque Value: 10 n•m [89 in-lb]



Last Modified: 11-Apr-2016